Experiment 4.4

Determination and plotting of Electric Field Intensity inside and outside sphere

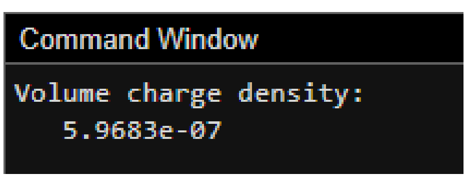
Code File Name: `Exp4_4_Sphere_EField.m`

Code

```
clc;
R = 2;
Q = 20e-6;
V = (4/3)*pi*R^3;
rho = Q/V;
disp("Volume charge density:");
disp(rho);
r = linspace(0.1,3,100);
k = 9e9;
E_inside = (k*rho.*r)/3;
E_outside = k*Q./(r.^2);
plot(r,E_inside,'LineWidth',2); hold on;
plot(r,E_outside,'LineWidth',2);
xlabel('Distance (m)');
ylabel('Electric Field (N/C)');
title('Electric Field inside and outside sphere');
legend('Inside','Outside');
```

Command Prompt

```
Volume charge density:
   5.9683e-07
```



Output Graph